Learning to Expand and Align: A Fourier-Based Vision–Language Framework for Incremental Hyperspectral Image Classification

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***Abstract*—With the rapid increase in satellite data acquisition, efficient adaptation to newly acquired hyperspectral scenes has become increasingly important. In this context, incremental learning provides a practical solution by enabling models to continuously learn new knowledge without retaining all historical data or performing full retraining. However, substantial distribution shifts across sensors, acquisition conditions, and imaging environments significantly limit the ability of existing methods to adapt from a labeled base scene to a newly acquired scene under few-shot supervision. Moreover, the cosine similarity-based prediction has difficulty capturing higher order statistical information and complex data relationships, which are important for improving classification performance. To address these challenges, we propose a Fourier-based vision–language framework for incremental hyperspectral image classification (HSIC), learning to expand and align (LEAF). In the base session, a Fourier-based domain expansion (FDE) strategy generates expanded domains to facilitate domain-invariant feature learning. To further enhance representation capability, a multilevel vision–language alignment module aligns visual and textual embeddings at both instance and class levels. During the inference stage, a covariance-enhanced prediction (CEP) strategy incorporates second-order statistical information to improve prediction robustness and reliability. Extensive experiments on three benchmark hyperspectral datasets demonstrate that the LEAF consistently achieves superior overall performance over state-of-the-art methods in incremental HSI classification (HSIC) tasks. The code will be available at https://github.com/ZhangYun123zy/TGRS_2026_LEAF**

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I. Introduction

HYPERSPECTRAL images (HSIs) acquired by high-resolution remote sensing (RS) sensors consist of hundreds of narrow spectral bands [1], [2], each capturing the continuous reflectance characteristics of ground objects [3], [4]. In HSI, every pixel corresponds to a spectral curve that represents the intrinsic reflectance property of the target material [5], [6]. Accurate classification of HSI is therefore significant for practical applications, including land-cover identification, environmental monitoring, and mineral resource exploration [7]. To achieve accurate HSI classification (HSIC), both traditional machine learning algorithms and deep learning models have been extensively investigated [8], [9]. Conventional methods typically rely on handcrafted features coupled with shallow classifiers, which often struggle to model the complex spectral–spatial correlations inherent in HSIs [10]. In contrast, deep learning techniques can automatically learn hierarchical feature representations directly from raw inputs, yielding substantial improvements in classification performance [11], [12]. Through successive architectural innovations, deep learning-based HSIC has achieved remarkable progress [13], [14].

In real-world HSIC applications, the high cost of pixel-level annotation makes it difficult to collect sufficient labeled samples for every newly acquired scene. A common strategy is to train a model on a labeled source domain (SD) and transfer it to a target domain (TD), which is generally referred to as cross-scene HSIC. However, substantial distribution shifts caused by variations in sensors, acquisition conditions, and imaging environments often lead to severe performance degradation when models are transferred across scenes [15]. To mitigate such shifts, unsupervised domain adaptation (UDA) methods align the distributions of labeled source data and unlabeled target data for target-scene classification [16]. When the target scene is unavailable during training, domain generalization (DG) further aims to learn domain-invariant representations from one or multiple SDs so that the model can generalize to a previously unseen domain [17], [18].

Despite their effectiveness in cross-scene transfer, UDA and DG methods are not fully aligned with the practical

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scenario where hyperspectral scenes are acquired sequentially over time. UDA typically assumes that the TD data are available during adaptation, whereas DG focuses on generalization to unseen domains without subsequent adaptation. In contrast, sequential cross-scene HSIC requires the model to continuously adapt to newly arriving domains while retaining knowledge acquired from previously observed scenes. With the rapid development of RS platforms, new HSIs are continuously collected from different sensors, regions, and acquisition conditions [19]. In such a sequential setting, previously used training data may be unavailable due to limited onboard storage, data transmission constraints, privacy policies, or security requirements. Meanwhile, the newly arrived scene usually contains only a few labeled samples per class because dense annotation remains expensive. Therefore, the model is expected to adapt to the new scene using limited supervision while preserving its performance on previously learned scenes. However, updating the model may bias the parameters toward the current domain, leading to the degradation of previously acquired knowledge. This phenomenon is known as catastrophic forgetting and becomes a major obstacle to sequential cross-scene HSIC.

In order to alleviate catastrophic forgetting, incremental learning has been proposed, which enables the system to continuously acquire new knowledge from new tasks while simultaneously retaining knowledge from the previously learned tasks. For instance, the learning without forgetting (LwF) model [20] only utilizes new data to train the network while retaining the original recognition capability. Besides, the elastic weight consolidation (EWC) method [21] preserves previous knowledge by constraining updates of parameters that are important to previous tasks. Zhao et al. [22] introduced an incremental learning strategy that continuously updates model weights using continuous spatial–spectral feature extraction. Li et al. [23] proposed a feature calibration and separation (FCS) method for nonexemplar class-incremental learning, approximating the drift of prototypes caused by model evolution.

However, faced with the challenge of a shift from one data source of HSI to another, pretrained deep learning networks often fail to accurately characterize the new dataset that has significant differences. Therefore, it is significant to mitigate the data distribution shifts caused by varying acquisition conditions. Overcoming this challenge is crucial for developing a robust, universal model capable of performing consistently across diverse datasets. Domain expansion provides an effective way to alleviate this problem by augmenting or generating source-related out-of-domain samples, thereby encouraging the model to learn more domain-invariant representations for transfer to the TD. For instance, Zhang et al. [24] pioneered the application of a single-source DG algorithm (i.e., SDEnet) to cross-scene HSIC, enabling the model to learn domain-invariant representations from synthesized extended domains. Besides, Zhao et al. [25] proposed a generation network for extension, which consists of a symmetric encoder–decoder to implicitly build local joint feature under style randomization. To address the discrepancy between SD and TDs, a spatial–spectral enhancement and causal constraint network (S2ECNet) in terms of both data enhancement and causal alignment is proposed in [26]. Specifically, a generator is created in S2ECNet to simulate the spectral deviation and spatial deviation from the TD, which makes up for the lack of data diversity in the SD.

Moreover, recent vision–language models represented by contrastive language-image pretraining (CLIP) [27] have opened up new possibilities for image classification and presented impressive understanding capabilities for RS tasks by leveraging language information and undergoing extensive contrastive training on large-scale text–image pairs. For instance, Hu et al. [28] leveraged the CLIP model to incorporate additional rich semantic priors, enriching the visual feature space. Specifically, they designed cross-modal alignment and cross-modal integration strategies to bridge the modal discrepancy between visual and textual representations. Although multiple studies successfully introduced language before HSIC tasks, none has explored the potential benefits of text information for incremental HSIC. To bridge this gap, it is crucial to investigate how vision–language frameworks can be leveraged to align cross-modal semantics and enhance domain representations, tackling the problems of model degradation and poor generalization when new hyperspectral datasets arrive.

Furthermore, during the inference stage, most existing methods utilize cosine similarity to measure the distance between sample features and class centers. Despite the simplicity, this isotropic approach struggles to capture higher order information and complex data relationships. To address this, anisotropic covariance metrics have garnered significant attention due to their ability to fully exploit statistical characteristics. For instance, Xi et al. [29] proposed a novel few-shot learning framework for HSIC that leverages a class-covariance metric, achieving state-of-the-art meta-task adapted classification with limited labeled samples. Additionally, Goswami et al. [30] demonstrated the superiority of modeling feature covariance relations within an optimal Bayes classifier, verifying its effectiveness over standard Euclidean metrics when learning from nonstationary data.

Driven by the aforementioned challenges, we develop the learning to expand and align (LEAF), a Fourier-based vision–language framework for incremental HSIC, which jointly performs domain expansion and modality alignment. In the base session, LEAF applies a random perturbation mechanism on the base data in the frequency space to generate expanded data. To characterize domain variations, we construct universal prototypes and domain-specific shift vectors. By jointly training on both base and expanded data, a generalized model is obtained for improved prediction across diverse datasets. Furthermore, we align the textual and visual embeddings at both instance and class levels, enhancing representation and generalization capability for HSIs. During the incremental domain training stage, the domain shift vectors are optimized to enhance classification performance. Subsequently, class probabilities are derived through cosine similarity-based probability calculation and covariance measurement.

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The main contributions of this work are as follows.

1) We propose a Fourier-based domain expansion (FDE) method for domain-invariant representation learning in incremental HSIC. By jointly optimizing the network on both the base and expanded domains, our approach yields a highly generalizable model capable of effectively mitigating data distribution shifts in subsequent sessions.
2) A multilevel vision–language alignment (MLVLA) module is devised, which can align text and visual embeddings at both the instance and class levels. By bridging these two levels of vision–language representation gaps, the model can establish a more accurate link between visual features and natural language for all classes.
3) To enhance prediction robustness, the covariance-enhanced prediction (CEP) strategy introduces an additional metric that maximizes the utilization of limited data, which balances prototype similarity with covariance-adjusted distributional affinity.
4) We thoroughly evaluate our framework across three HSI scenes, and extensive experimental results demonstrate its superiority over the existing methods.

The remainder of this article is organized as follows. Section II briefly introduces the related works on CLIP models and incremental learning. Section III describes our proposed LEAF framework in detail. Section IV carries out abundant experiments and analyzes the classification results. Section V draws the conclusion.

II. Related Works

A. CLIP Models

Vision–language pretrained models like CLIP establish precise cross-modal correspondences through extensive contrastive training on text–image pairs, exhibiting remarkable transferability across various downstream vision and language tasks [31]. Recently, vision–language pretrained models have demonstrated strong representation and understanding capabilities in RS tasks.

For example, Qin et al. [32] implemented a multimodal prototype correction strategy that leverages pretrained language models to extract spatial and spectral text features from spatial and spectral text descriptions, thereby aligning image spatial prototypes with spatial text features and spectral prototypes with spectral text features to enhance the recognition of diverse ground objects. Furthermore, Pang et al. [33] developed a zero-shot HSIC framework based on CLIP, aiming to eliminate the reliance on manual annotations. To improve recognition accuracy (AA) for objects of various sizes, they also proposed a resolution scaling strategy that fuses predictions from multiple image resolutions, enhancing the quality of pseudo-labels generated by CLIP.

The remarkable success of CLIP in the RS tasks inspires us to explore its potential application to incremental learning for HSIC. Despite the promising advances of CLIP in various RS tasks, its adaptation to the incremental learning paradigm in HSI remains underexplored. To address this gap, we propose the MLVLA to align text and visual embeddings at both the instance and class levels, yielding a generalizable model.

B. Incremental Learning

Incremental learning is a powerful technique that allows models to progressively learn new tasks or identify new classes in changing environments. Instead of starting training from the beginning each time, incremental learning updates the model step-by-step, effectively preventing the loss of previously acquired knowledge, known as catastrophic forgetting [34].

For example, Bai et al. [35] proposed an incremental learning method based on a geometric spatial–spectral feature integrated network and class-incremental learning framework, enabling the model to adapt to new class data without excessively forgetting knowledge of old classes during training. To bridge the recognition gap and safeguard prior knowledge during incremental learning, Yu et al. [36] introduced a distillation-constrained prototype representation network (DCPRN), which couples feature-level and decision-level knowledge transfer to guide updates while maintaining old class information. In [37], the proposed method incorporates a min-max cross selection module tailored to HSI characteristics with a cross spectral–spatial knowledge distillation module. Besides, Li et al. [38] devised HyperSC, a novel framework that synergizes spectral-consistent auxiliary sample synthesis with stability–plasticity fused learning. Faced with the plasticity–stability dilemma, Jiang et al. [39] proposed a novel network architecture that integrates contrastive learning and an angular penalty loss. Synergistically, these components ensure robust knowledge retention, transfer, and adaptability.

Although these methods have achieved strong classification performance and significantly mitigated catastrophic forgetting, they overlook a critical limitation: the cross-scene generalization capability for the same categories across different domains. Specifically, existing approaches fail to explicitly address performance degradation when models encounter identical semantic classes but under varying environmental conditions, sensor configurations, or geographic distributions. This domain shift problem remains inadequately explored in the context of incremental HSIC.

III. Proposed Leaf for HSIC

The overall architecture of the proposed LEAF is illustrated in Fig. 1. The training procedure comprises two stages: base domain training and incremental domain training. In the base session, we apply the FDE to the original data to generate expanded domains. Data from both the base and expanded domains are fed into the image encoder for joint network training. Universal prototypes are adapted via domain-specific shift vectors to obtain domain-specific prototypes. Coarse- and fine-grained text features are fused to complement each other, yielding semantically rich representations. To align visual and textual representations, we propose the MLVLA to optimize the network from both instance and class perspectives. In the incremental session, domain shift vectors are optimized to enhance classification performance. Class probabilities are subsequently derived through cosine similarity-based probability calculation and covariance measurement.

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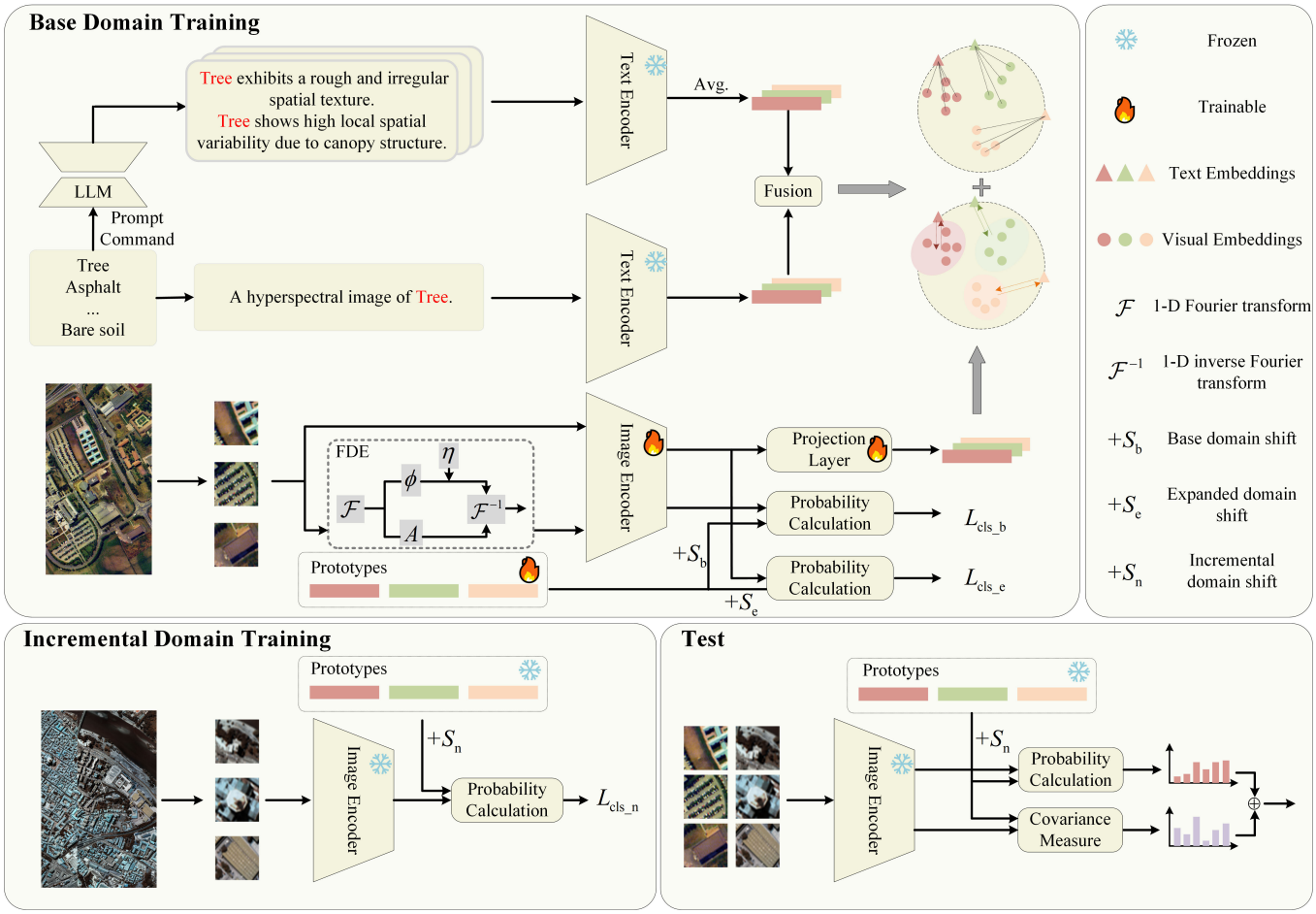


Fig. 1. Overall architecture of the proposed LEAF for incremental HSIC. The training procedure comprises two stages: base domain training and incremental domain training. During the training stage, we introduce the FDE and the MLVLA to obtain a more generalizable model. During the test stage, class probabilities are derived through probability calculation and covariance measurement.

A. Fourier-Based Domain Expansion

To effectively capture the high-dimensional characteristics of HSIs and accommodate their unique data acquisition properties, we construct an expanded domain during the base training stage, enabling the model to generalize under distribution shifts across different sessions.

The Fourier transform maps an HSI patch into the frequency domain, providing a convenient way to introduce controlled perturbations for simulating distribution variations. The resulting Fourier representation is characterized by amplitude and phase components. Instead of assuming a strict semantic-style decomposition, we use phase perturbation as a controlled spatial-frequency-domain augmentation strategy. The amplitude spectrum is kept unchanged to preserve the dominant spatial-frequency energy distribution of each spectral band, while the phase spectrum is slightly perturbed to simulate domain-dependent spatial-frequency variations within local HSI patches.

Given an input HSI patch $\mathbf{X} \in \mathbb{R}^{L\times H\times W}$, where L, H, and W denote the number of spectral bands, patch height, and patch width, respectively, we process each spectral band independently. For the lth spectral band, its spatial patch is flattened into a 1-D sequence

$$\mathbf{x}_l = \left[x_{l,0}, x_{l,1}, \ldots, x_{l,M-1}\right] \in \mathbb{R}^M, \quad M = H \times W. \tag{1}$$

The 1-D Fourier transform is then applied to the flattened spatial sequence

$$X_{l,k} = \sum_{m=0}^{M-1} x_{l,m} \exp\left(-j2\pi \frac{mk}{M}\right), \quad k = 0, \ldots, M-1. \tag{2}$$

Each Fourier coefficient $X_{l,k}$ can be represented in polar form

$$X_{l,k} = A_{l,k} \exp\left(j\phi_{l,k}\right) \tag{3}$$

where $A_{l,k} = |X_{l,k}|$ and $\phi_{l,k} = \arg(X_{l,k})$ denote the amplitude and phase components of the kth frequency coefficient in the lth spectral band, respectively.

To construct an expanded domain, we preserve the amplitude spectrum and perturb the phase spectrum with Laplace-distributed noise

$$\hat{\phi}_{l,k} = \phi_{l,k} + \rho\eta_{l,k} \tag{4}$$

where

$$\eta_{l,k} \sim \text{Laplace}\left(0, \frac{1}{|\varepsilon|}\right). \tag{5}$$

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Here ρ and ε control the perturbation strength and the scale of the Laplace noise, respectively. In this way, the phase perturbation introduces controlled spatial-frequency variations while maintaining the dominant amplitude distribution of the original HSI patch.

The perturbed Fourier coefficient is reconstructed as

$$\hat{X}_{l,k} = A_{l,k} \exp\left(j\hat{\phi}_{l,k}\right). \tag{6}$$

And the expanded-domain sequence is obtained through the inverse 1-D Fourier transform

$$\hat{\mathbf{x}}_l = \mathcal{F}^{-1}\left(\hat{\mathbf{X}}_l\right). \tag{7}$$

The reconstructed sequence $\hat{\mathbf{x}}_l$ is reshaped back to the spatial size $H \times W$. By applying this operation to all spectral bands, we obtain the expanded HSI patch $\hat{\mathbf{X}} \in \mathbb{R}^{L\times H\times W}$.

The FDE expands the base domain distribution through controlled perturbations, encouraging the model to learn representations that are more transferable across domains. Nevertheless, such expansion cannot fully account for the complex discrepancies among real HSI scenes. Therefore, domain-specific shift vectors are further introduced to capture the residual domain-dependent variations that are not sufficiently represented by the shared feature space. Based on this complementary design, each class representation is decomposed into a universal component and a domain-specific shift component.

Let d and C denote the embedding dimension and the number of land-cover classes, respectively. The universal class prototypes are represented by

$$\mathbf{P} = \left[\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_C\right]^\top \in \mathbb{R}^{C\times d} \tag{8}$$

where $\mathbf{p}_c$ denotes the domain-shared prototype of class c.

For each domain q, we further introduce a domain-specific shift matrix

$$\mathbf{S}^q = \left[\mathbf{s}_1^q, \mathbf{s}_2^q, \ldots, \mathbf{s}_C^q\right]^\top \in \mathbb{R}^{C\times d} \tag{9}$$

where $\mathbf{s}_c^q \in \mathbb{R}^d$ denotes the shift vector of class c in domain q. In this work, $q = b$, $q = e$, and $q = n$ represent the base domain, the expanded domain, and the novel domain, respectively.

The domain-specific prototype of class c in domain q is then defined as

$$\boldsymbol{\mu}_c^q = \frac{\mathbf{p}_c}{\|\mathbf{p}_c\|_2} + \delta\frac{\mathbf{s}_c^q}{\|\mathbf{s}_c^q\|_2} \tag{10}$$

where δ controls the contribution of the domain-specific shift component. This formulation allows the model to preserve domain-shared class semantics through $\mathbf{p}_c$ while adapting to domain-specific variations through $\mathbf{s}_c^q$.

During base training, the image encoder, projection layer, universal prototypes $\mathbf{P}$, base domain shift matrix $\mathbf{S}^b$, and expanded-domain shift matrix $\mathbf{S}^e$ are jointly optimized. During the incremental session, only the newly introduced domain-specific shift matrix $\mathbf{S}^n$ is optimized using the few labeled samples from the current domain, while all other model parameters and previously learned domain-specific shift matrices remain fixed. At test time, no explicit domain identity is provided to the model, and all test samples are classified using the prototypes $\boldsymbol{\mu}_n^q$ obtained after the current incremental adaptation. This design enables the model to adapt to the novel domain with limited trainable parameters while preserving the decision space of previously learned domains.

B. Multilevel Vision–Language Alignment

In its standard usage, CLIP relies on simple prompt templates (e.g., "a photograph of a [CLS].") to construct zero-shot classifiers for visual tasks. However, this category-agnostic approach has inherent limitations. The generic nature of these templates fails to capture category-specific nuances, making them particularly inadequate for fine-grained classification tasks where detailed feature discrimination is crucial. A single, static template cannot effectively characterize the diverse visual characteristics within each category, especially when subtle distinctions between categories are essential.

According to the prior knowledge of land-cover classes, coarse-grained and fine-grained text descriptions are assigned to each land-cover class. Concretely, "an HSI of [CLS]" is used as a template to construct coarse-grained text descriptions for each class in a cloze way. As for fine-grained text, we leverage large language models (LLMs) to generate dynamic, category-specific descriptions from the perspective of spatial and spectral characteristics. These LLM-generated descriptions provide richer, more discriminative semantic representations for each category, enhancing the model's capacity to capture fine-grained features. To encode these textual descriptions, a lightweight text encoder built upon the Transformer architecture is employed. The text encoder consists of three Transformer layers with eight attention heads and contains approximately 33 M parameters.

After obtaining the coarse-grained text features f_{coarse} and fine-grained text features f_{fine}, we generate the integrated text features f_{text} as follows:

$$f_{\text{text}} = (1-\beta)\, f_{\text{coarse}} + \beta f_{\text{fine}} \tag{11}$$

where β is a coefficient to fuse the two text features.

At the instance level, the MLVLA establishes precise correspondences between individual visual samples and their textual descriptions. At the class level, the MLVLA captures structural correspondences between category-level visual clusters and their semantic textual prototypes.

Given a pair of visual–text embeddings $\{f_i^V, f_i^T\}$, an image classifier for textual label set $\mathcal{T}_C$ can be implemented by computing the posterior probabilities of classes via

$$\mathcal{L}_{\text{In}}^V = \frac{1}{n}\sum_{i=1}^{n}\sum_{t_j\in\mathcal{T}_C} -\mathbb{I}\left(t_j = t_i\right)\log p\left(t_j|\mathbf{x}_i; \mathcal{T}_C\right). \tag{12}$$

Similarly, the corresponding loss function for texts is

$$\mathcal{L}_{\text{In}}^T = \frac{1}{n}\sum_{i=1}^{n}\sum_{x_j\in\mathcal{D}_{\text{tr}}} -\mathbb{I}\left(\mathbf{x}_j = \mathbf{x}_i\right)\log p\left(\mathbf{x}_j \mid t_i; \mathcal{D}_{\text{tr}}\right) \tag{13}$$

where $\mathcal{D}_{\text{tr}}$ is the training dataset.

The instance-level objective function is

$$\mathcal{L}_{\text{In}} = \frac{\mathcal{L}_{\text{In}}^V + \mathcal{L}_{\text{In}}^T}{2}. \tag{14}$$

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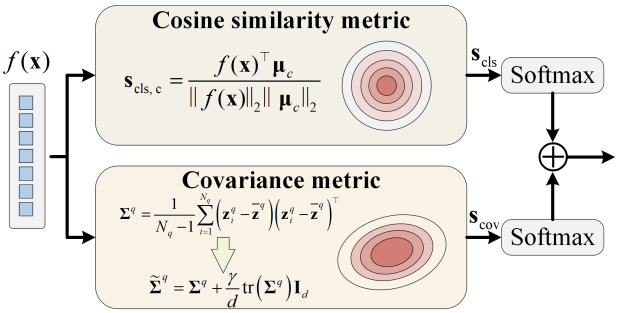


Fig. 2. Overview of the CEP strategy, which combines cosine similarity-based probability calculation and covariance measurement.

To reduce the distance between category clusters and their corresponding textual descriptions, class-level contrastive learning is proposed. By minimizing the distance between textual embeddings and category means, the model achieves class-level alignment

$$\mathcal{L}_{\text{Cl}}^V = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\cos\left(P_i^V, f_i^T\right)\right)}{\sum_{j=1}^{N}\exp\left(\cos\left(P_i^V, f_j^T\right)\right)} \tag{15}$$

where P_i^V is the mean visual feature vector of the ith class and N denotes the number of classes. L_{Cl}^T is computed in a similar manner to L_{Cl}^V

$$\mathcal{L}_{\text{Cl}}^T = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\cos\left(f_i^T, P_i^V\right)\right)}{\sum_{j=1}^{N}\exp\left(\cos\left(f_i^T, P_j^V\right)\right)}. \tag{16}$$

The final class-level alignment loss L_{Cl} is represented as follows:

$$\mathcal{L}_{\text{Cl}} = \frac{\mathcal{L}_{\text{Cl}}^V + \mathcal{L}_{\text{Cl}}^T}{2}. \tag{17}$$

The overall loss for base training is summarized as follows:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{cls_b}} + \omega_1\mathcal{L}_{\text{cls_e}} + \omega_2\mathcal{L}_{\text{In}} + \omega_3\mathcal{L}_{\text{Cl}} \tag{18}$$

where $\mathcal{L}_{\text{cls_b}}$ and $\mathcal{L}_{\text{cls_e}}$ represent the classification cross-entropy losses for the base and expanded domains, respectively. Additionally, $\mathcal{L}_{\text{In}}$ and $\mathcal{L}_{\text{Cl}}$ denote the instance-level and class-level contrastive losses. The weighting factors ω_1–ω_3 are assigned to $L_{\text{cls_e}}$, L_{In}, and L_{Cl}.

C. Covariance-Enhanced Prediction

In Section III-B, the proposed method successfully integrates linguistic and visual information while mitigating modal bias. Among them, a widely used isotropic measure is the cosine similarity between the sample feature $f(\mathbf{x})$ and the class prototype $\boldsymbol{\mu}_c$, which is defined as

$$\mathbf{s}_{\text{cls, c}} = \frac{f(\mathbf{x})^\top \boldsymbol{\mu}_c}{\|f(\mathbf{x})\|_2 \cdot \|\boldsymbol{\mu}_c\|_2}. \tag{19}$$

Although this metric is straightforward and effective, it essentially remains a distance measure centered around the distribution mean. Such isotropic approaches, despite their simplicity, often fail to capture higher order statistical information and complex intrinsic relationships within the data. To overcome these limitations, we propose to leverage statistical information from available data for more comprehensive and effective evaluation, and introduce an anisotropic metric that can reflect the shape of data distribution. Class probabilities are derived through cosine similarity-based probability calculation and covariance measurement. Fig. 2 illustrates the flowchart of the CEP strategy.

For each evaluation domain q, we estimate a domain-specific covariance matrix from the visual features of the available samples in that domain. Let $\mathcal{D}_q^{test} = \{\mathbf{x}_i^q\}_{i=1}^{N_q}$ the test set of domain q, where N_q is the number of test samples. The corresponding visual embeddings and mean feature vector are, respectively, defined as $\mathbf{z}_i^q = f(\mathbf{x}_i^q) \in \mathbb{R}^d$ and $\bar{\mathbf{z}}^q = 1/N_q \sum_{i=1}^{N_q}\mathbf{z}_i^q$.

Then, the covariance matrix of domain q is estimated by

$$\boldsymbol{\Sigma}^q = \frac{1}{N_q - 1}\sum_{i=1}^{N_q}\left(\mathbf{z}_i^q - \bar{\mathbf{z}}^q\right)\left(\mathbf{z}_i^q - \bar{\mathbf{z}}^q\right)^\top. \tag{20}$$

To avoid matrix singularity and improve numerical stability, we apply trace-based regularization

$$\tilde{\boldsymbol{\Sigma}}^q = \boldsymbol{\Sigma}^q + \frac{\gamma}{d}\text{tr}\left(\boldsymbol{\Sigma}^q\right)\mathbf{I}_d \tag{21}$$

where γ is a hyperparameter controlling the regularization strength, d denotes the feature dimension, $\mathbf{I}_d$ is the identity matrix, and tr$(\cdot)$ represents the trace operator.

The covariance matrix captures the global feature dispersion and anisotropic structure of the corresponding evaluation domain, providing complementary statistical information to the prototype-based cosine similarity. For a given test sample $\mathbf{x}$, we compute its score relative to category c as

$$\mathbf{s}_{\text{cov,c}}(\mathbf{x}) = -\frac{1}{d}\left(f(\mathbf{x}) - \boldsymbol{\mu}_c^n\right)^T \tilde{\Sigma}^{-1}\left(f(\mathbf{x}) - \boldsymbol{\mu}_c^n\right). \tag{22}$$

The division by feature dimension d serves as a normalization factor, preventing the softmax transformation into probabilities from collapsing into a single category. The negative sign maintains metric consistency, ensuring that a larger output score indicates a higher probability of the sample belonging to category c.

We then normalize the scores of two metrics using the softmax function: $\mathbf{P}_{\text{cls}} = \text{softmax}(\mathbf{s}_{\text{cls}})$, and $\mathbf{P}_{\text{cov}} = \text{softmax}(\mathbf{s}_{\text{cov}})$. The final probability score $\mathbf{P}$ is obtained via a weighted combination of the prototype-based scores and the visual covariance metric-based scores

$$\mathbf{P} = \alpha\mathbf{P}_{\text{cls}} + (1 - \alpha)\mathbf{P}_{\text{cov}}. \tag{23}$$

The final prediction of this sample is determined by $\arg\max_c \mathbf{P}_c$. This integrated approach enables the model to balance prototype similarity with covariance-adjusted distributional affinity, leading to more robust and discriminative classification predictions.

IV. Experimental Results and Analysis

A. Data Description and Evaluation Measures

1) Houston Dataset: The Houston13 (HU2013) scene was captured in June 2012 with a CASI-1500 sensor over the University of Houston and surrounding urban area. The Houston18 (HU2018) was acquired in February 2017 at 1-m spatial

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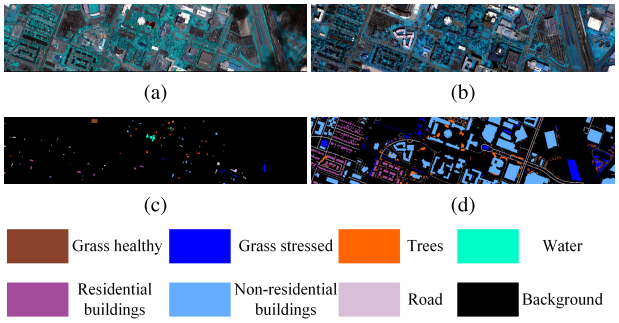


Fig. 3. Houston dataset visualization. (a) Houston2013 false-color image. (b) Houston2018 false-color image. (c) Houston2013 ground-truth map. (d) Houston2018 ground-truth map.

TABLE I
SAMPLE DISTRIBUTION OF HOUSTON DATASET SCENES HOUSTON13 AND HOUSTON18

Class	Name	Houston13	Houston18
C1	Grass healthy	345	1353
C2	Grass stressed	365	4888
C3	Trees	365	2766
C4	Water	285	22
C5	Residential buildings	319	5347
C6	Non-residential buildings	408	32459
C7	Road	443	6365
Total		**2530**	**53200**

TABLE II
SAMPLE DISTRIBUTION OF PAVIA DATASET SCENES PAVIAU AND PAVIAC

Class	Name	PaviaU	PaviaC
C1	Tree	3064	7598
C2	Asphalt	6631	9248
C3	Brick	3682	2685
C4	Bitumen	1330	7287
C5	Shadow	947	2863
C6	Meadow	18649	3090
C7	Bare soil	5029	6584
Total		**39332**	**39355**

resolution with 48 spectral bands. To spatially and spectrally align with the HU2018 dataset, a 209×955 subregion was extracted from HU2013, and its spectral channels were adjusted to 48 bands. Although the two collections cover the same wavelength range, their substantial temporal and spatial differences induce a marked domain shift. The task involves seven land-cover categories, as shown in Table I. The SD contains 2530 samples, whereas the TD contains 53 200 samples. Fig. 3 shows the false-color images and ground-truth maps for both datasets.

2) Pavia Dataset: The University of Pavia (PaviaU) and the Pavia Center (PaviaC) are contained in the Pavia dataset. Reflective Optics Spectrographic Image System (ROSIS), which has spectral coverage from 430 to 860 nm, was used to acquire these scenes. The PaviaU includes 103 spectral bands, 610×340 pixels, and a spatial resolution of 1.3 m. The PaviaC has 102 bands and 1096×715 pixels. The last band of PaviaU is removed to ensure consistency with PaviaC in terms of the number of spectral bands. Their labels are the same and provided in Table II. Their false-color images and ground-truth maps are shown in Fig. 4.

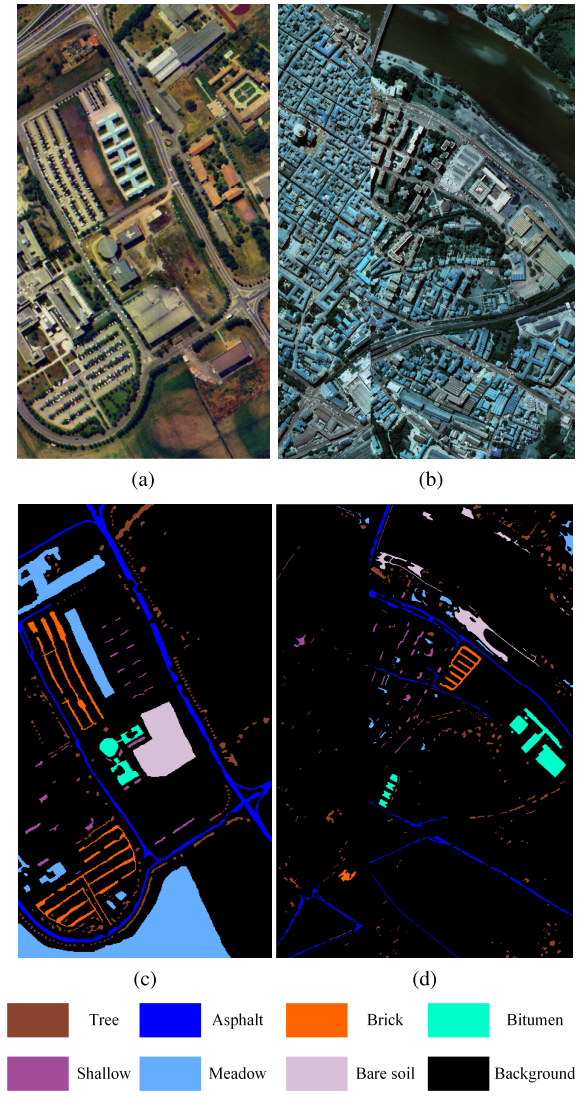


Fig. 4. Pavia dataset visualization. (a) PaviaU false-color image. (b) PaviaC false-color image. (c) PaviaU ground-truth map. (d) PaviaC ground-truth map.

TABLE III
SAMPLE DISTRIBUTION OF OHID DATASET SCENES OHID1 AND OHID2

Class	Name	OHID1	OHID2
C1	Building	19163	45895
C2	Farmland	85615	99264
C3	Road	21997	9942
C4	Water	116637	61004
C5	Fishpond	1892	6638
Total		**245304**	**222743**

3) OHID Dataset: The OHID dataset is collected by "Zhuhai No.1" hyperspectral satellite (OHS) constellation and designed and produced by Zhuhai Orbita Aerospace Technology Company Ltd. (Orbita). It comprises two HSIs sourced

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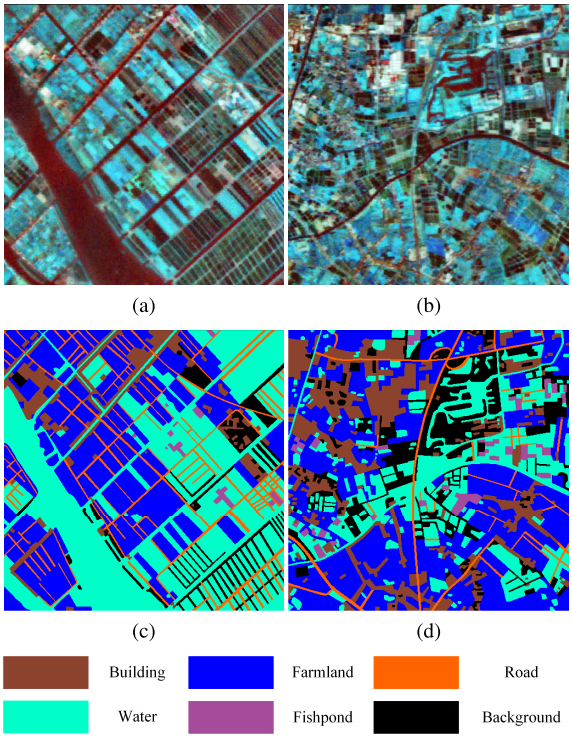


Fig. 5. OHID dataset visualization. (a) OHID1 false-color image. (b) OHID2 false-color image. (c) OHID1 ground-truth map. (d) OHID2 ground-truth map.

from diverse regions of Zhuhai City, China, each boasting 32 spectral bands with a spatial resolution of 10 m and spanning a spectral range of 400–1000 nm [40]. Both subsets include five classes as shown in Table III. Fig. 5 shows the false-color images and ground-truth maps for both datasets.

To evaluate the performance, we utilize standard metrics in HSIC, including mean accuracy (AA), overall accuracy (OA), and the kappa coefficient (κ). Additionally, in the context of incremental learning, we propose the last-session accuracy (LSA), which quantifies the model's performance across all n datasets. LSA can be formulated as follows:

$$\text{LSA} = \frac{1}{n}\sum_{i=1}^{n} \text{OA}_i. \tag{24}$$

B. Experimental Settings

We conduct extensive experiments to evaluate and compare the classification performance of several incremental learning approaches for HSI analysis. In addition to the proposed method, we include a range of representative baselines, including Finetuning, LwF [20], EWC [21], FCS [23], HyperKD [37], HyperSC [38], and LLL-AFCA [39]. For a fair comparison, the implementation settings and parameters of these methods are primarily adopted from their original studies and further fine-tuned to achieve optimal performance under our experimental conditions. All experiments are independently repeated ten times with different random seeds.

For base domain training, mini-batches from the base dataset are fed into the network for 100 epochs. Learning rates are set to 0.05 for three tasks. For the base domain, we split the Houston dataset with 30% labeled samples for training,

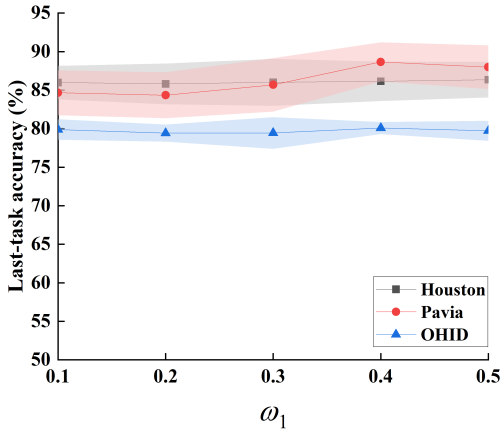


Fig. 6. Impact of ω_1 on LSA for the Houston, Pavia, and OHID tasks.

and split the Pavia and OHID dataset with 10% labeled samples for training. The batch size is 64, with patch sizes of 13×13 for Houston, Pavia, and OHID. GPT-4 is employed to generate category-specific fine-grained text descriptions from both spatial-texture and spectral-reflectance perspectives. Each land-cover class is associated with ten descriptions, consisting of five spatial-texture descriptions and five spectral-reflectance descriptions, which are subsequently used to construct the corresponding fine-grained textual representation. Specifically, we evaluate different values of the weight parameter ω_1 from the set $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. As shown in Fig. 6, the LSA for Houston, Pavia, and OHID reaches its peak when ω_1 is set to 0.5, 0.4, and 0.4, respectively. This observation suggests that the cross-entropy loss computed on the expanded domain plays a crucial role in improving the generalization capability of the model under distribution shifts across different sessions. To investigate the impact on the classification performance of ω_2 and ω_3, we carry out experiments under the condition of $\omega_2 \in \{0.01, 0.02, 0.03, 0.04\}$ and $\omega_3 \in \{0.1, 0.2, 0.3, 0.4\}$. The classification performance of the three datasets is shown in Fig. 7. From Fig. 7, we choose the optimal parameters $\{0.01, 0.1\}$, $\{0.03, 0.1\}$, and $\{0.01, 0.1\}$ for Houston, Pavia, and OHID, respectively. It can be observed that appropriate values of ω_2 and ω_3 consistently improve classification performance, indicating that the two weighting parameters are essential for balancing the contributions of L_{In} and L_{Cl} at the instance and class levels, respectively. Furthermore, a parameter sensitivity analysis is performed to identify the optimal value of β from the candidate set $0.1, 0.2, 0.3, 0.4, 0.5, 0.6$. Fig. 8 shows that the best performance is achieved when β is set to 0.4, 0.2, and 0.2 for Houston, Pavia, and OHID, respectively. Since β controls the relative contribution of coarse-grained and fine-grained text features, its value directly affects the semantic representation of the integrated text feature f_{text}. Specifically, coarse-grained text features provide category-level semantic guidance, whereas fine-grained text features offer more discriminative cues for separating similar classes. The results indicate that an appropriate balance between these two types of textual information is essential for robust classification. When β is excessively large, the model may overemphasize fine-grained descriptions and weaken global semantic consistency.

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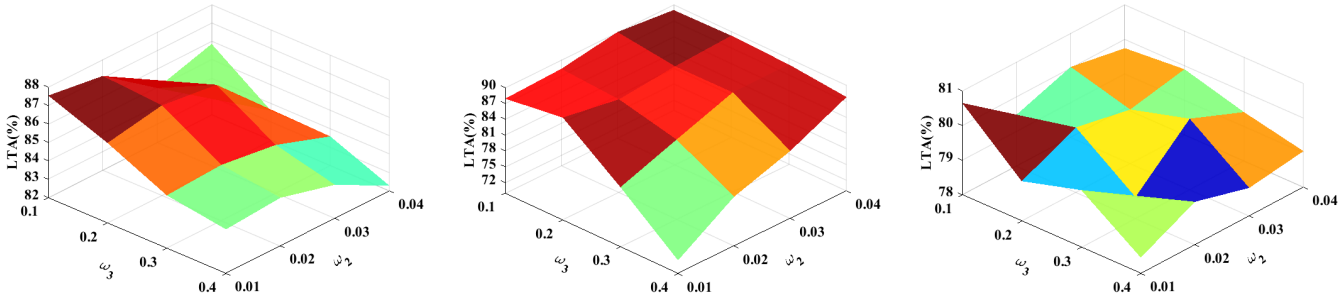


Fig. 7. LSA (%) of various methods on the three datasets with different ω_2 and ω_3. (a) Houston. (b) Pavia. (c) OHID.

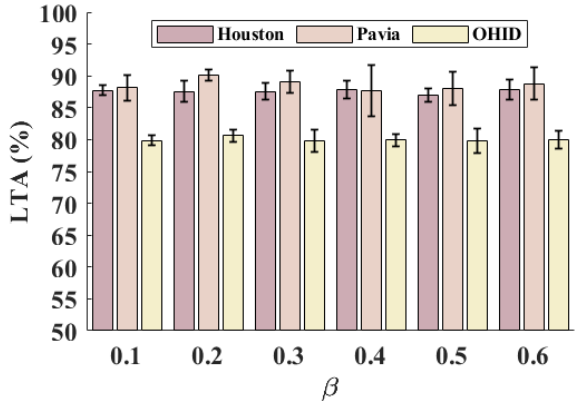


Fig. 8. Impact of β on LSA for the Houston, Pavia, and OHID tasks.

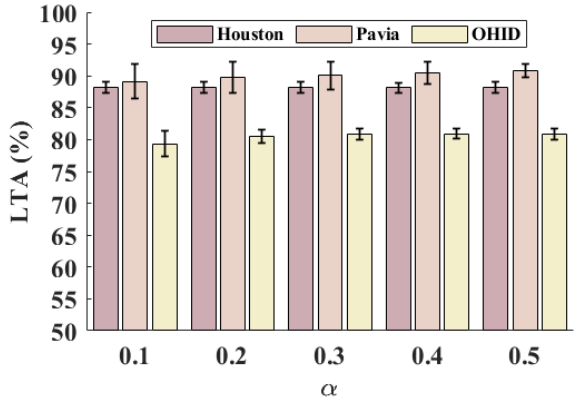


Fig. 9. Impact of the weighting factor α on LSA for the Houston, Pavia, and OHID tasks.

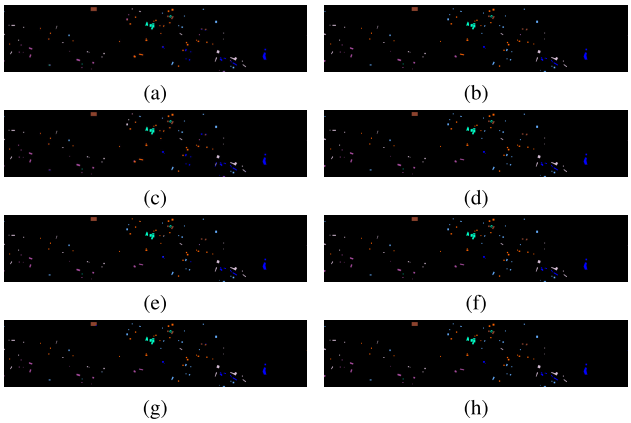


Fig. 10. Classification maps obtained via various algorithms for the Houston dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

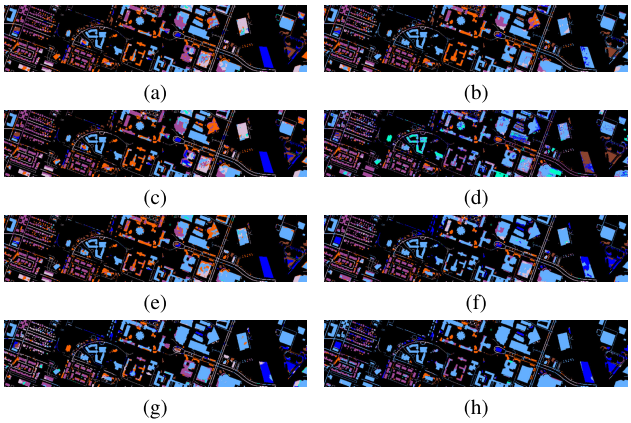


Fig. 11. Classification maps obtained via various algorithms for the Houston18 dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

Conversely, overly small values of β may limit the contribution of discriminative fine-grained cues, thereby reducing the separability of similar land-cover categories.

In the incremental stage, each class is limited to five labeled samples, following a standard few-shot setting. The domain shift vectors are trained for 500, 500, and 350 epochs for Houston, Pavia, and OHID, respectively, with a learning rate of 0.01. The coefficient δ is set to 0.01 to control the contribution of the domain-specific shift vectors during prototype adaptation, enabling moderate adjustment to domain-specific variations while preserving the shared class semantics encoded by the universal prototypes. We evaluate the impact of the weighting factor α by varying it from 0.1, 0.2, 0.3, 0.4, and 0.5. As illustrated in Fig. 9, the optimal values of the weighting factor α for the Houston, Pavia, and OHID datasets are 0.1, 0.5, and 0.4, respectively. It implies that the appropriate value of α balances the contributions of the prototype-based scores

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TABLE IV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HOUSTON DATASET

Dataset	No	Finetuning	LwF [20]	EWC [21]	FCS [23]	HyperKD [37]	HyperSC [38]	LLL-AFCA [39]	Ours
HU2013	1	72.31 ± 34.29	98.68 ± 2.86	69.13 ± 33.57	85.83 ± 19.59	99.24 ± 1.04	**100.00 ± 0.00**	98.35 ± 2.34	99.34 ± 1.72
	2	72.12 ± 33.49	99.29 ± 0.99	68.55 ± 37.33	87.81 ± 25.96	94.11 ± 15.89	**99.81 ± 0.20**	99.14 ± 1.21	99.37 ± 0.66
	3	80.00 ± 22.22	**100.00 ± 0.00**	86.77 ± 11.11	97.14 ± 4.68	99.86 ± 0.31	**100.00 ± 0.00**	97.96 ± 1.57	99.81 ± 0.31
	4	98.15 ± 3.38	87.35 ± 22.15	90.37 ± 18.98	88.89 ± 14.06	87.59 ± 28.18	**100.00 ± 0.00**	99.40 ± 1.14	99.90 ± 0.30
	5	73.27 ± 21.04	96.50 ± 5.47	77.35 ± 24.31	92.15 ± 14.43	98.94 ± 3.18	**99.86 ± 0.40**	98.21 ± 2.01	98.43 ± 3.01
	6	79.09 ± 14.54	99.23 ± 0.96	73.64 ± 14.11	89.30 ± 13.70	97.33 ± 3.96	**99.79 ± 0.63**	96.33 ± 3.02	97.83 ± 3.16
	7	86.77 ± 24.57	99.81 ± 0.39	84.94 ± 22.88	94.45 ± 13.06	99.94 ± 0.11	**100.00 ± 0.00**	99.19 ± 1.49	**100.00 ± 0.00**
	AA	80.25 ± 9.42	97.27 ± 3.30	78.68 ± 9.10	90.80 ± 8.98	96.72 ± 4.95	**99.92 ± 0.09**	98.37 ± 0.62	99.24 ± 0.87
	OA	80.05 ± 9.70	97.69 ± 2.66	78.51 ± 9.18	90.95 ± 9.15	97.06 ± 4.24	**99.92 ± 0.10**	98.33 ± 0.63	99.23 ± 0.88
	Kappa	76.65 ± 11.34	97.29 ± 3.12	74.85 ± 10.72	89.41 ± 10.72	96.54 ± 4.98	**99.91 ± 0.12**	98.04 ± 0.74	99.10 ± 1.03
HU2018	1	73.78 ± 15.32	74.87 ± 6.60	72.62 ± 13.88	57.38 ± 12.78	72.31 ± 9.17	**75.97 ± 8.79**	65.87 ± 16.72	60.06 ± 11.98
	2	63.50 ± 13.23	21.05 ± 12.04	65.85 ± 11.22	38.16 ± 21.62	22.17 ± 15.44	64.83 ± 7.80	60.31 ± 12.81	**75.28 ± 5.12**
	3	67.87 ± 10.35	62.01 ± 14.02	67.07 ± 8.93	53.46 ± 8.27	68.11 ± 12.79	**70.53 ± 3.84**	56.02 ± 15.45	57.43 ± 8.13
	4	**100.00 ± 0.00**	70.59 ± 31.35	**100.00 ± 0.00**	68.23 ± 34.32	63.53 ± 36.90	**100.00 ± 0.00**	97.65 ± 7.06	90.59 ± 9.56
	5	72.43 ± 12.55	69.06 ± 8.95	76.21 ± 11.58	56.42 ± 9.59	64.43 ± 12.36	**79.38 ± 6.63**	69.14 ± 13.59	60.90 ± 12.22
	6	68.25 ± 12.09	75.79 ± 6.48	65.71 ± 13.85	48.59 ± 12.04	80.38 ± 4.49	78.66 ± 3.86	59.28 ± 9.45	**87.52 ± 5.05**
	7	**65.45 ± 10.99**	45.05 ± 11.26	64.66 ± 10.53	39.07 ± 16.75	41.69 ± 12.21	53.17 ± 7.55	36.43 ± 12.55	50.76 ± 11.93
	AA	73.04 ± 1.86	59.77 ± 5.09	73.16 ± 2.01	51.62 ± 8.63	58.94 ± 8.61	**74.65 ± 1.73**	63.53 ± 4.26	68.93 ± 3.35
	OA	68.03 ± 6.15	65.67 ± 4.23	66.91 ± 7.10	47.76 ± 8.64	67.95 ± 2.98	73.93 ± 1.23	57.64 ± 6.63	**77.07 ± 2.12**
	Kappa	53.46 ± 6.02	45.31 ± 5.84	52.47 ± 6.80	30.73 ± 6.82	47.27 ± 4.92	59.46 ± 1.46	40.80 ± 6.71	**60.99 ± 2.92**
	LSA	74.04 ± 4.39	81.68 ± 2.43	72.71 ± 4.53	69.36 ± 5.74	82.50 ± 2.56	86.92 ± 0.61	77.98 ± 3.44	**88.15 ± 0.85**

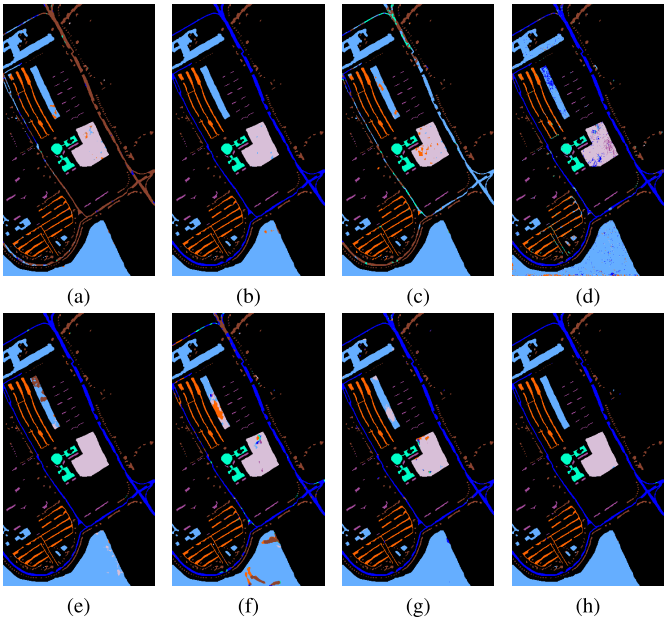


Fig. 12. Classification maps obtained via various algorithms for PaviaU dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

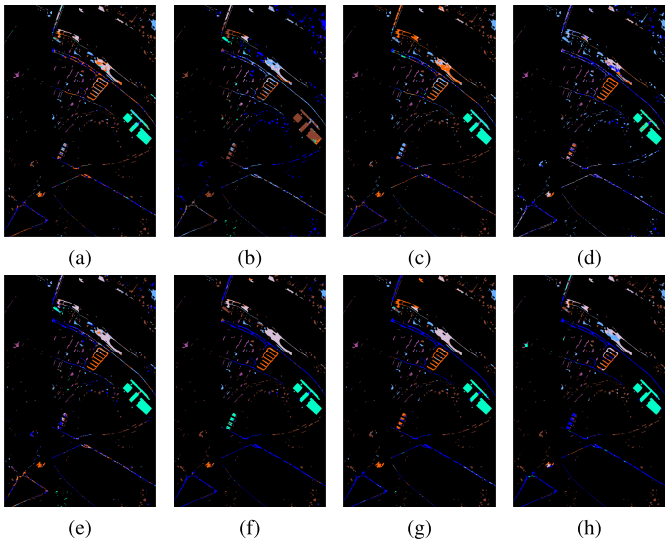


Fig. 13. Classification maps obtained via various algorithms for PaviaC dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

and the visual Mahalanobis metric-based scores, leading to more robust and discriminative classification predictions.

C. Comparing With Other Methods

The experimental results of various incremental learning-based methods are summarized in Tables IV–VI, with the top-performing outcomes highlighted in bold. For visual comparison, we exhibit the best classification maps of various methods on the datasets from Figs. 10 to 15.

Table IV summarizes the classification performance of various methods on the Houston dataset. The proposed LEAF method achieves the highest overall performance, surpassing the second-best method by 1.23% in LSA. Additionally, the LEAF exhibits the second-lowest variance across multiple independent runs, underscoring its stability and robustness. The visualization results, which are shown in Figs. 10 and 11, further verify that our method maintains a leading edge in two scenes and show outstanding classification performance in areas marked in blue. Table V presents the classification performances of different methods on the Pavia task. HyperSC, which integrates spectral-consistent auxiliary sample synthesis with progressive enhancement and fusion distillation, and LLL-AFCA, which adopts multilevel alignment strategies, perform outstandingly in LSA. Notably, compared to other methods, our method attains higher metrics, achieving an LSA improvement of 1.65%, and exhibits smaller deviations across multiple experiments, highlighting the robustness of

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TABLE V

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE PAVIA DATASET

Dataset	No	Finetuning	LwF [20]	EWC [21]	FCS [23]	HyperKD [37]	HyperSC [38]	LLL-AFCA [39]	Ours
PaviaU	1	83.81 ± 9.13	97.24 ± 1.57	91.43 ± 5.09	85.50 ± 15.94	97.58 ± 1.54	96.16 ± 1.53	98.01 ± 1.55	**98.99 ± 0.48**
	2	71.00 ± 8.09	**99.16 ± 0.86**	71.32 ± 13.74	81.62 ± 11.46	99.04 ± 1.07	81.77 ± 24.02	96.81 ± 2.90	98.78 ± 1.31
	3	76.40 ± 21.98	95.61 ± 5.23	85.48 ± 16.59	78.43 ± 13.71	97.01 ± 3.14	96.31 ± 3.26	**97.63 ± 1.78**	96.53 ± 4.55
	4	88.64 ± 22.72	85.10 ± 23.53	83.24 ± 28.98	88.58 ± 13.31	87.18 ± 20.09	99.29 ± 1.24	**99.35 ± 0.96**	96.67 ± 3.15
	5	87.98 ± 9.40	98.03 ± 2.37	84.55 ± 12.15	94.16 ± 7.00	98.59 ± 1.62	98.90 ± 1.44	**99.61 ± 0.53**	99.59 ± 0.50
	6	87.69 ± 12.66	96.94 ± 2.50	82.43 ± 17.49	70.42 ± 35.51	96.54 ± 3.38	88.89 ± 3.36	97.03 ± 2.15	**99.48 ± 0.64**
	7	64.03 ± 21.88	84.29 ± 18.42	70.85 ± 20.43	42.35 ± 35.81	89.95 ± 16.26	96.13 ± 1.54	95.84 ± 4.41	**98.66 ± 2.76**
	AA	79.94 ± 6.06	93.77 ± 5.07	81.33 ± 7.17	77.29 ± 8.44	95.13 ± 3.96	93.92 ± 4.55	97.75 ± 1.07	**98.39 ± 1.04**
	OA	80.53 ± 5.05	95.22 ± 2.49	80.14 ± 8.71	71.83 ± 18.05	95.98 ± 2.13	90.47 ± 5.76	97.11 ± 1.50	**98.85 ± 0.80**
	Kappa	73.15 ± 6.16	93.24 ± 3.61	73.28 ± 10.69	64.59 ± 19.24	94.35 ± 3.05	87.10 ± 7.52	95.99 ± 2.06	**98.39 ± 1.13**
PaviaC	1	90.05 ± 9.59	**95.38 ± 2.06**	91.50 ± 6.67	33.62 ± 21.17	95.04 ± 2.74	77.84 ± 9.89	74.57 ± 8.30	91.05 ± 4.95
	2	93.48 ± 4.05	**96.29 ± 2.60**	94.08 ± 4.54	57.64 ± 18.87	95.90 ± 2.35	88.36 ± 7.94	81.01 ± 10.42	86.57 ± 3.59
	3	77.21 ± 13.18	42.28 ± 18.91	81.93 ± 11.03	62.28 ± 18.99	40.13 ± 20.29	**94.54 ± 4.20**	86.79 ± 9.87	44.34 ± 22.95
	4	75.58 ± 14.03	31.82 ± 20.64	67.39 ± 21.23	67.30 ± 27.56	31.18 ± 21.98	**89.62 ± 5.66**	82.21 ± 6.20	84.11 ± 2.53
	5	98.56 ± 1.45	99.59 ± 0.63	98.16 ± 1.67	71.88 ± 36.66	**99.74 ± 0.33**	98.59 ± 2.24	97.91 ± 2.26	90.05 ± 5.61
	6	**90.05 ± 8.54**	87.43 ± 6.49	82.13 ± 18.09	65.18 ± 18.22	86.53 ± 7.48	84.39 ± 10.04	86.76 ± 6.60	72.16 ± 5.75
	7	78.33 ± 16.25	85.77 ± 9.81	83.16 ± 9.48	48.42 ± 13.84	**85.93 ± 6.45**	80.14 ± 6.91	75.49 ± 10.13	83.55 ± 6.44
	AA	86.18 ± 3.74	76.94 ± 4.20	85.48 ± 3.10	58.04 ± 12.91	76.35 ± 4.45	**87.64 ± 2.64**	83.53 ± 2.21	78.83 ± 3.41
	OA	85.96 ± 3.88	78.28 ± 3.78	85.34 ± 3.64	55.19 ± 11.08	77.82 ± 3.84	**86.04 ± 3.90**	81.14 ± 2.58	82.72 ± 1.76
	Kappa	**83.16 ± 4.65**	73.60 ± 4.69	82.46 ± 4.24	46.80 ± 13.00	73.03 ± 4.77	83.27 ± 4.62	77.50 ± 3.03	79.15 ± 2.17
	LSA	83.25 ± 3.63	86.75 ± 2.59	82.74 ± 5.43	63.51 ± 11.27	86.90 ± 2.04	88.25 ± 2.00	89.13 ± 1.49	**90.78 ± 1.04**

TABLE VI

CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE OHID DATASET

Dataset	No	Finetuning	LwF [20]	EWC [21]	FCS [23]	HyperKD [37]	HyperSC [38]	LLL-AFCA [39]	Ours
OHID1	1	67.40 ± 12.69	77.93 ± 6.35	69.00 ± 13.83	78.36 ± 12.02	77.91 ± 6.02	32.78 ± 5.57	**80.71 ± 4.55**	74.12 ± 7.06
	2	59.94 ± 23.14	88.28 ± 5.78	78.03 ± 10.51	88.35 ± 3.16	89.48 ± 3.61	77.44 ± 5.12	75.81 ± 3.14	**92.55 ± 3.23**
	3	36.79 ± 13.32	36.94 ± 11.49	42.15 ± 14.80	53.04 ± 8.77	37.56 ± 9.31	16.54 ± 7.07	**70.03 ± 8.50**	14.88 ± 5.51
	4	68.80 ± 23.30	93.99 ± 3.10	78.67 ± 17.34	92.53 ± 2.42	94.48 ± 2.42	86.63 ± 4.70	78.68 ± 5.04	**96.37 ± 1.73**
	5	79.91 ± 20.69	73.36 ± 18.52	85.33 ± 18.38	90.65 ± 1.96	75.56 ± 17.45	88.74 ± 3.54	**94.43 ± 1.79**	64.04 ± 18.93
	AA	62.57 ± 8.06	74.10 ± 4.53	70.64 ± 7.44	**80.59 ± 3.16**	75.00 ± 4.03	60.43 ± 0.90	79.93 ± 2.11	68.39 ± 5.43
	OA	62.82 ± 15.75	85.47 ± 1.32	74.47 ± 8.65	**86.41 ± 1.76**	86.19 ± 1.43	72.94 ± 2.29	77.19 ± 2.66	85.74 ± 0.84
	Kappa	46.58 ± 20.34	76.92 ± 2.00	62.50 ± 10.77	**78.83 ± 2.75**	78.05 ± 2.10	58.63 ± 2.54	66.60 ± 3.56	76.72 ± 1.51
OHID2	1	48.90 ± 13.04	51.79 ± 7.38	46.56 ± 11.97	49.35 ± 12.89	51.38 ± 6.82	41.48 ± 8.83	44.10 ± 9.58	**57.90 ± 7.18**
	2	45.39 ± 20.88	76.83 ± 5.30	68.42 ± 10.73	70.44 ± 8.73	77.52 ± 3.95	71.41 ± 10.32	57.85 ± 7.20	**88.34 ± 3.62**
	3	27.81 ± 13.14	25.70 ± 8.00	33.67 ± 14.21	31.34 ± 10.23	27.43 ± 9.90	**38.01 ± 12.81**	36.76 ± 4.88	5.75 ± 4.13
	4	55.44 ± 22.73	84.82 ± 4.67	59.77 ± 20.27	77.61 ± 7.26	84.96 ± 4.37	82.15 ± 5.57	67.56 ± 16.87	**86.84 ± 3.37**
	5	56.58 ± 22.02	28.25 ± 9.99	56.50 ± 22.19	**79.56 ± 4.02**	30.03 ± 13.02	71.68 ± 6.67	50.47 ± 13.72	25.14 ± 16.95
	AA	46.82 ± 9.08	53.48 ± 2.58	52.99 ± 8.19	**61.66 ± 1.54**	54.26 ± 3.69	60.95 ± 1.61	51.35 ± 3.64	52.79 ± 3.86
	OA	48.42 ± 11.02	70.13 ± 1.86	59.64 ± 7.97	66.59 ± 2.77	70.52 ± 1.41	66.70 ± 4.89	56.52 ± 4.91	**76.09 ± 1.11**
	Kappa	31.60 ± 13.41	56.70 ± 2.43	44.57 ± 9.73	53.16 ± 3.23	57.26 ± 1.89	54.43 ± 5.07	41.75 ± 5.80	**63.64 ± 1.93**
	LSA	55.62 ± 12.67	77.80 ± 1.02	67.05 ± 8.01	76.50 ± 1.81	78.36 ± 1.03	69.82 ± 3.33	66.85 ± 2.87	**80.92 ± 0.83**

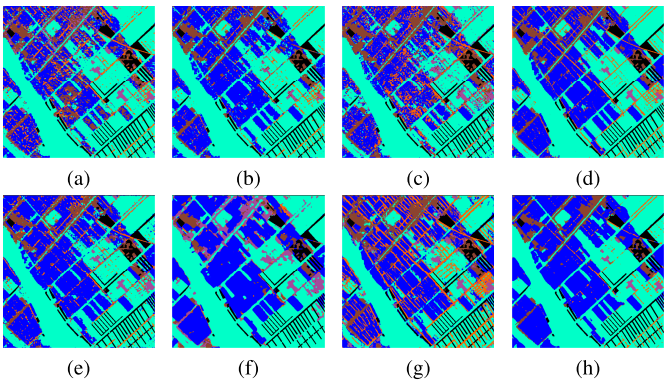


Fig. 14. Classification maps obtained via various algorithms for OHID1 dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

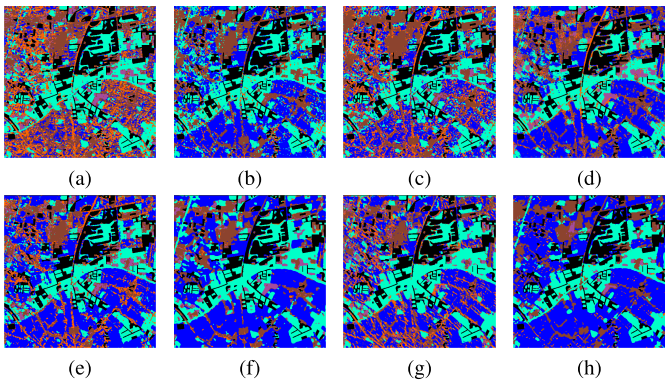


Fig. 15. Classification maps obtained via various algorithms for OHID2 dataset. (a) Finetuning. (b) LwF. (c) EWC. (d) FCS. (e) HyperKD. (f) HyperSC. (g) LLL-AFCA. (h) LEAF.

model. Figs. 12 and 13 indicate that our model exhibits lower noise levels and closely aligns with the ground truth. Table VI presents the classification performances of different methods on the OHID dataset. The LEAF achieves the highest

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TABLE VII

CLASSIFICATION PERFORMANCE OF THE ABLATIONS ON THE PROPOSED LEAF ON HOUSTON, PAVIA, AND OHID

Dataset	Case	FDE	MLVLA	CEP	Base	Inc	LSA
Houston	1				95.83 ± 5.78	73.66 ± 3.20	84.74 ± 2.97
	2	✓			98.15 ± 4.51	74.58 ± 1.72	86.36 ± 2.31
	3	✓	✓		99.23 ± 0.65	76.50 ± 2.29	87.87 ± 1.40
	4	✓	✓	✓	**99.24 ± 0.88**	**77.05 ± 2.13**	**88.15 ± 0.85**
Pavia	1				97.52 ± 2.37	72.03 ± 6.29	84.78 ± 3.41
	2	✓			98.71 ± 0.57	78.62 ± 4.72	88.66 ± 2.54
	3	✓	✓		**98.90 ± 0.77**	81.32 ± 1.52	90.11 ± 0.92
	4	✓	✓	✓	98.85 ± 0.80	**82.72 ± 1.76**	**90.78 ± 1.04**
OHID	1				84.88 ± 1.12	74.41 ± 1.51	79.65 ± 1.10
	2	✓			85.73 ± 1.02	74.44 ± 1.22	80.08 ± 0.78
	3	✓	✓		**85.80 ± 0.79**	75.50 ± 1.34	80.65 ± 0.96
	4	✓	✓	✓	85.74 ± 0.84	**76.09 ± 1.11**	**80.92 ± 0.83**

LSA of 80.92%, demonstrating the best overall performance across the two OHID scenes. Nevertheless, relatively low accuracies are observed for Road and Fishpond, which may be attributed to their challenging spectral–spatial characteristics. In particular, the 10-m spatial resolution of OHID may cause narrow road regions to be mixed with surrounding land-cover types, while the Fishpond class contains considerably fewer samples in the base domain and may exhibit spectral similarity to Water. Despite these challenging categories, LEAF achieves competitive performance on classes such as Farmland and Water, with relatively low standard deviations, showing its overall robustness to spectral variations across OHID scenes. In Figs. 14 and 15, the classification maps of LEAF exhibit higher visual consistency with the ground-truth maps for "Farmland" and "Water." This observation is also in agreement with the quantitative results reported in Table VI.

D. Ablations on the LEAF

As described in Section III, the proposed LEAF mainly comprises the FDE to learn domain-invariant representation, the MLVLA to align text and visual embeddings at both the instance and class levels, and the CEP to enhance prediction robustness. To verify the effectiveness of each module, we conduct ablation studies primarily on the LEAF component in this section.

In case 1, the proposed network excludes FDE, MLVLA, and CEP. In case 2, the proposed network excludes MLVLA and CEP. In case 3, the proposed network excludes CEP. In case 4, the complete experimental setup encompasses three primary contributions.

The average classification performance and standard deviation across three experimental datasets are presented in Table VII. It is observed that the progressive performance improvement with increasingly complete configurations reflects the cumulative effectiveness of the proposed modular components in incremental learning. The design of the FDE enables the model to achieve better generalization, capable of handling data distribution shifts in subsequent sessions. The MLVLA aligns text and visual embeddings at both the instance and class levels. At the instance level, precise correspondences are established between individual visual samples and their textual descriptions. At the class level, structural correspondences are captured between category-level visual clusters and their semantic textual prototypes. By reducing the instance- and class-level vision–language representation gaps, the model will establish a more accurate link between visual features and natural language for all classes. To enhance predictive robustness, the CEP introduces an additional metric designed to maximize the utilization of limited data. This module balances prototype similarity with covariance-adjusted distributional affinity.

TABLE VIII

BASE TRAINING TIME, INCREMENTAL TRAINING TIME, AND TEST TIME (SECONDS) OF VARIOUS METHODS ON EACH DATASET

Methods	Houston			Pavia			OHID		
	base.	inc.	test	base.	inc.	test	base.	inc.	test
Finetuning	15.31	2.64	23.01	96.45	3.64	44.08	608.52	3.45	82.05
LwF	17.85	3.87	22.59	92.58	3.64	44.09	613.94	2.66	88.43
EWC	23.18	6.62	23.27	102.14	6.08	47.37	612.38	5.79	93.51
FCS	221.72	248.39	13.69	697.59	276.88	16.78	3294.89	1208.63	85.67
HyperKD	20.01	4.47	22.22	103.41	4.46	43.67	575.08	4.71	89.98
HyperSC	24.98	160.24	7.40	130.24	150.28	10.13	738.49	250.38	55.70
LLL-AFCA	39.70	64.77	11.44	104.81	138.71	18.11	584.94	592.33	53.36
Ours	100.29	5.33	8.12	762.22	5.28	13.15	3018.41	3.09	57.12

E. Analysis of the Computational Complexity

To evaluate the practical applicability of the LEAF in real-world scenarios, we conduct a comprehensive analysis of its computational efficiency. The evaluation focuses on comparing the training time of base session, the training time of incremental session, and the inference time in the incremental session of the LEAF with mainstream incremental algorithms, including Finetuning, LwF, EWC, FCS, HyperKD, HyperSC, and LLL-AFCA. All methods are implemented using the NVIDIA GeForce RTX 4090 GPU, leveraging the open-source deep learning framework PyTorch. The results in Table VIII show that LEAF requires a relatively longer base training time compared to most competing methods. However, its incremental training and test times remain at a moderate level, indicating its capability to efficiently learn new knowledge from incremental datasets while maintaining fast prediction speed. This trade-off suggests that the LEAF prioritizes more thorough representation learning during the base training phase, which in turn facilitates efficient adaptation in subsequent stages. Overall, the LEAF strikes a tradeoff between computational efficiency and classification performance, demonstrating its practicality for real-world applications.

V. CONCLUSION

In this article, we propose LEAF to learn to expand and align for incremental HSIC. Concretely, the LEAF employs FDE for domain-invariant representation learning, which achieves a generalized model capable of handling data distribution shifts in subsequent sessions by jointly training on both base data and expanded data. Moreover, the MLVLA module aligns visual and textual embeddings at both instance and class levels to further enhance feature representation. In the test stage, we propose the CEP for probability prediction, which leverages both the cosine similarity between sample features and class prototypes, as well as statistical information

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from the available data, to enable more comprehensive and reliable evaluation. Experiments on Houston, Pavia, and OHID datasets show superior results of the LEAF over existing incremental methods in LSA, confirming the efficacy of our strategy against catastrophic forgetting. Although our method has achieved success in incremental learning under cross-scene conditions, the base training phase remains time-consuming. In future work, we will focus on reducing the computational cost of base training to further improve the overall efficiency of the framework.

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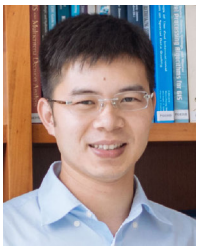
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